Cross-disciplinary applications of the Toulbar2 solver

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Abstract

The **valued constraint satisfaction problem** (VCSP) or **weighted constraint satisfaction problem** (WCSP) is a generalization of the classic **CSP**. We try to find an assignment of minimal cost to a set of variables given **soft** and **hard constraints**. It is usually solved using a **branch-and-bound search** (BnB).

By finding a good **lower bound** to a given instance, subtrees of the BnB can be quickly pruned to efficiently find an optimal solution. This paper studies the cross-disciplinary application of the solver **Toulbar2**, ranging from solving difficult combinatorial problems, to real-world application such as bioinformatics and resource management and finally artificial intelligence. In many cases where the problem was able to be written using the VCSP framework, **Toulbar2 was able to outperform** other state-of-the-art solvers.

Combinatorial problem solving

Cluster Editing for Graphs

Zahn's decision problem can be described as the cluster editing problem. Given a graph, what is the minimal number of operations (adding edges, removing edges) needed to create a **disjoint union of cliques**. Below an exemple of the cluster editing problem:

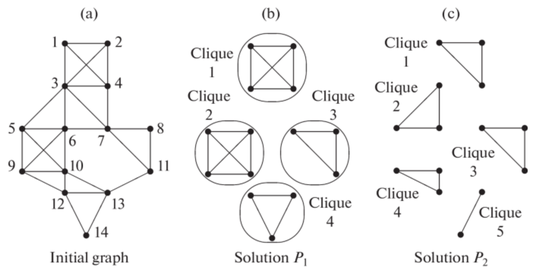


Fig. 1: Example of SMC: Robust Supply Chain with Different Network Size.[3]

By transforming the cluster editing problem into a **VCSP** and solving it using Toulbar2, Mhamdi and Naanaa (2014) obtained **results much better** than the fixed parameter algorithm for both sparse and dense graphs.

A Complicated Chess Problem

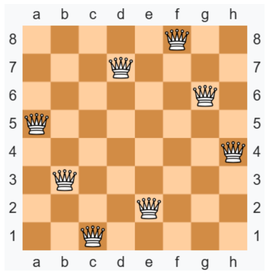


Fig. 2: 8-Queens example [5]

The N-Queen problem is a **non-trivial mathematical problem** that looks for a placement of N queens on a chessboard of size N by N where no two queens are threatening each other. The figure to the left shows an example of a solution where $n = 8$. The weighted N-Queen problem is the difficult optimization problem of minimizing the cost of the N-Queen if every tile incurs a cost. By modeling the **weighted N-Queen problem** as a VCSP, Sewa et al. (2022) demonstrated that **Toulbar2 significantly outperforms** other state-of-the-art solvers.

Other problems such as the quadratic assignement problem, the **knapsack problem with conflict graph and discrete integrations** were also solved using the VCSP framework and showed **good performance of the Toulbar2** solver.

Bioinformatics & Life Sciences

Computational Protein Design (CPD)

CPD aims to create **artificial proteins** by identifying the amino acid sequence that minimizes the protein's energy, known as the **Global Minimum Energy Conformation** (GMEC). This is an NP-hard optimization problem due to the vast search space created by the numerous possible side-chain conformations, called rotamers.

Modeling with CFNs and Toulbar2

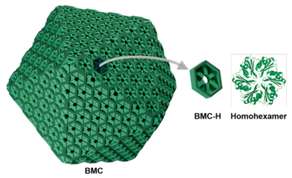


Fig. 3: BMC-H homohexamer of bacterial microcompartment (BMC) shell.[2]

To solve this, the protein structure is mapped into a **CFN**: residues become variables, rotamers become values, and physical interactions are encoded as costs. Toulbar2 uses advanced **branch-and-bound and pruning techniques** not only to find the exact **GMEC**, but also to enumerate near-optimal solutions to analyze the complex fitness landscape of the protein.

Hybrid Deep Learning Approaches

Recently, hybrid approaches have emerged to tackle multi-state design. By combining **deep learning models** (which produce scoring functions) with **Toulbar2's** exact optimization under explicit constraints, researchers can successfully design complex, symmetrical multi-component proteins—like the homohexamer shown above while simultaneously avoiding undesirable negative states.

Operational Research & Planning

Spatio-Temporal Allocation Problem

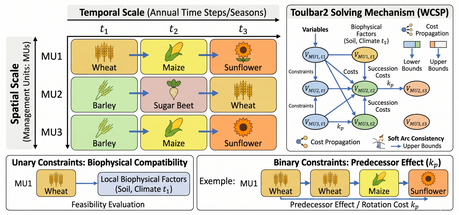


Fig. 4: Spatio-temporal crop rotation (Inspired by [1]).

The **crop allocation problem** requires assigning crop types to agricultural plots over time.

It utilizes the **WCSP** framework to balance **hard constraints** (e.g., agronomic feasibility and soil properties) with **soft constraints** (e.g., farmer preferences and soil balance).

While **WCSP** offers excellent modeling flexibility, **Toulbar2** faces **strong competition** from traditional **Mixed-Integer Linear Programming** (MILP) solvers when scaling up to very large farms.

Human Resources & Global Constraints

Hospital staff scheduling presents a different combinatorial challenge: balancing **strict coverage requirements** with individual nurse rest preferences.

Historically, **WCSPs** struggled here because modeling human rules required **exponentially large networks** of binary constraints. To solve this, **Toulbar2** integrated soft global constraints, allowing the solver to **evaluate a nurse's entire weekly schedule** as a single evaluation.

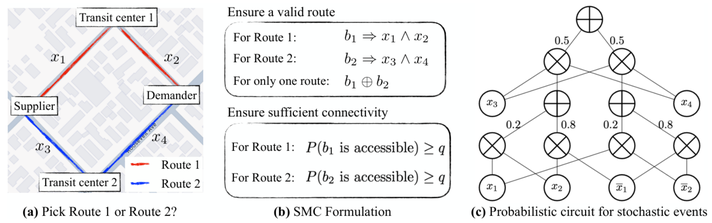
This leap in expressivity allows **Toulbar2** to **successfully compete** with traditional Constraint Programming solvers in human scheduling.

Hybridization for Dynamic Problems

Toulbar2 excels in deterministic planning, but its efficiency depends heavily on the **problem's topology**. For example, in **dynamic air traffic conflict** resolution, **Toulbar2** handles multiple variables well but struggles when the domain size (possible trajectories) grows too large. In such complex industrial applications, Toulbar2 achieves the **best response times** when hybridized with fast, non-optimal greedy heuristics to prune bad solutions early.

Artificial Intelligence & Machine Learning

Toulbar2 struggles with real world applications since exhaustive search creates computational bottlenecks on large instances. We can integrate **machine learning**, introducing new ways to improve the solution quality. For example, algorithms like **KOCO SMC** handle large instances by **tracking probability bounds** during the inference process. This early evaluation prevents the solver from performing exhaustive enumerations.



(a) Pick Route 1 or Route 2? **(b)** SMC Formulation **(c)** Probabilistic circuit for stochastic events

Fig. 5: Example of SMC: Robust Supply Chain with Different Network Size.[4]

Deep learning models can also transform raw sensory data directly into Graphical Models. This integration merges the perception of neural networks with the exact reasoning of constraint solvers to **increase overall flexibility**. For example, a neural network can translate unmapped protein structures or raw images into cost matrices, allowing Toulbar2 to focus exclusively on executing the **exact discrete optimization**.

Conclusion

The objective of this state-of-the-art review was to understand how a solver based on CFNs, such as Toulbar2, established itself across such diverse disciplines. Toulbar2 was shown to be a **highly pratical and versatile solver**, capable of tackling a wide variety of real-world and theoritcal problems within the cost function network framework. By leveraging CFNs, it is able to model complex systems ranging from biological structures to scheduling and combinatorial optimization. Furthermore, Toulbar2 is able to obtain optimal solutions within competitive computational times, often outperforming other state-of-the-art solvers. Toulbar2 clearly demonstrates the **strenght of CFN framework**.

However, its limitations on very large-scale instances hightlight the inherent cost of exact optimization. In this context, hybrid approaches that combine exact solving with learning-based methods or heuristics emerge as a promising solution. Overall, even though the solver Toulbar2 has its **limitation**, it has shown to be a relevant and good solver today, especially when **paired with other algorithms**.

References

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